Yunhan Qiao

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Ph.D. Candidate, Computer Science, Oregon State University

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Research Interests

My research sits at the intersection of empirical software engineering, code comprehension, and human–AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

Education

Oregon State University — Corvallis, OR, United States
Ph.D. in Computer Science, School of EECS — *Sep 2023 – Jun 2027 (expected)*
Advisor: Dr. Christopher Hundhausen at the SEAL Lab
Research Areas: Empirical Software Engineering, Code Comprehension, Computing Education, Human–AI Interaction

Arizona State University — Tempe, AZ, United States
M.S. in Software Engineering — *Aug 2021 – May 2023*
Advisor: Dr. Robert LiKamWa
Thesis: Networked System for Volumetric Athletic Coaching in Augmented Reality

Guangxi University of Science and Technology — Liuzhou, China
B.E. in Software Engineering — *Sep 2016 – Jun 2020*

Southern Cross University — Gold Coast, Australia
B.E. in Information Technology — *Sep 2017 – Jun 2020*

Publications (Google Scholar Page)

Peer-Reviewed Journal Articles

[J2] Tariq, A., Conrad, P., Hundhausen, C., Chandrasekaran, S., **Qiao, Y.**, & Haque, S. (2026). Using SlackBot-prompted reflections to promote contribution equality in team software development projects. *ACM Transactions on Computing Education.* https://doi.org/10.1145/3857221

[J1] **Qiao, Y.**, Shihab, M. I. H., & Hundhausen, C. (2026). A systematic literature review of the use of GenAI assistants for code comprehension: Implications for computing education research and practice. *ACM Transactions on Computing Education, 26*(2), Article 24, 1–33. https://doi.org/10.1145/3785366
Selected by the TOCE editorial board for presentation at SIGCSE TS 2027.

Peer-Reviewed Conference Proceedings

[C2] Motwani, M., Kulkarni, A., **Qiao, Y.**, Davis, M., & Chen, Z. (2025). LLM-guided differential fuzzing for detecting platform-specific bugs in scientific applications. *2025 IEEE Conference on AI x Software Engineering (AIxSE)*, 59–66. https://doi.org/10.1109/AIxSE64906.2025.00015

[C1] Shihab, M. I. H., Hundhausen, C., Tariq, A., Haque, S., **Qiao, Y.**, & Mulanda, B. W. (2025). The effects of GitHub Copilot on computing students' programming effectiveness, efficiency, and processes in brownfield coding tasks. *Proceedings of the 2025 ACM Conference on International Computing Education Research (ICER '25), 1*, 407–420. https://doi.org/10.1145/3702652.3744219

Under Review

[R2] **Qiao, Y.**, Haque, S., & Hundhausen, C. (2026). Beyond the prompt: Linking what developers ask, do, and understand with coding agents. Under review at *IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2027).* https://arxiv.org/abs/2609.33712

[R1] **Qiao, Y.**, Shihab, M. I. H., Haque, S., & Hundhausen, C. (2025). Faster but not wiser: GitHub Copilot decouples programming performance from code comprehension in brownfield tasks. Under review at *ACM Transactions on Computing Education.* https://arxiv.org/abs/2511.02922

Under Submission

[S1] Schutz, O., **Qiao, Y.**, Moussaoui, F. A., Choudhuri, R., & Sarma, A. (2026). Heart and mind in the loop: Why we vibe when we work with AI. Submitted to *International Conference on Cooperative and Human Aspects of Software Engineering (CHASE 2027).*

Teaching Experience

Graduate Teaching Assistant, School of EECS — Oregon State University
Undergraduate software engineering courses — *2024 – current*

Courses: CS 362 Software Engineering II; CS 461 Senior Software Engineering Project; CS 467 Online Capstone Project; CS 492 Mobile Software Development

Professional Experience

Software Engineering Intern, Mocha Software Nanjing, China
Software development *Aug 2020 – Nov 2020*
Developed a conference system for China Mobile Migu Culture & Technology Group with Spring, Spring MVC, and MyBatis, using Ajax to cut page rendering time from 1 s to 0.5 s and dynamic SQL with Redis-backed SSO sessions to reduce search load.

Awards & Scholarships

Oregon State University Scholarly Presentation Award *2026*
Oregon State University EECS Graduate Scholarship *2023 – current*
Guangxi University of Science and Technology Merit Award *2018*

Academic Service

Program Committee
ACM Technical Symposium on Computer Science Education (SIGCSE TS), 2027
IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2027

Reviewer
ACM Transactions on Computing Education (TOCE)

Skills & Coursework

Research Methods: Qualitative Analysis, Quantitative Analysis, Interviews, Survey Design
Programming: Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
Frameworks & Tools: React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Figma; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
Coursework: Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis

Languages

English (Fluent), Mandarin Chinese (Native)